

VLA学习与项目复现

1. 发展历程

- 规则时代：清晰的代码定义行为
- 端到端模仿学习
- 世界模型和强化学习
- VLA (Vision-Language-Action) :理解自然语言指令，看懂视觉信息，生成动作序列

要区别于VLM模型，Vision-Language Model的输出是语言相关信息：如描述、回答问题、生成文本表示。

2. 宇树 UnifoLM-VLA-0 模型

- 项目官网 - github: <https://unigen-x.github.io/unifolm-vla.github.io/>
- Code - github: <https://github.com/unitreerobotics/unifolm-vla>
- model - Huggingface: <https://huggingface.co/unitreerobotics/models>
- datasets - Huggingface: <https://huggingface.co/collections/unitreerobotics/unifolm-vla-0>

模型基于开源的Qwen-2.5 VL-7B，有强大的语义理解能力，通过340h的真机数据进行训练，**增强空间感知能力**。集成了动作分块预测技术，并引入了前向和逆向动力学约束，可以在执行动作前，模型就对动作序列进行了建模和预演。用**单一策略网络 (Checkpoint)** 高质量完成**12类任务**，优于OpenVLA。

3. OpenVLA 复现



为了复现UnifoLM-VLA-0，先复现OpenVLA练手

- Model <https://huggingface.co/openvla/openvla-7b>
- 微调Model: <https://huggingface.co/openvla/openvla-7b-finetuned-libero-spatial>
- 项目 <https://github.com/openvla/openvla>
- LIBERO环境 <https://github.com/Lifelong-Robot-Learning/LIBERO.git>

目标：复现LIBERO仿真基线中libero-spatial、libero-object、libero-goal、libero-10中的效果。

3.1 在gpulab平台租用服务器

(做仿真实验要租带图形化桌面的服务器)

服务器规格：RTX3090 1*24G显存36TFLOps 16核64G Intel_CPU



3.2 云服务器环境测试

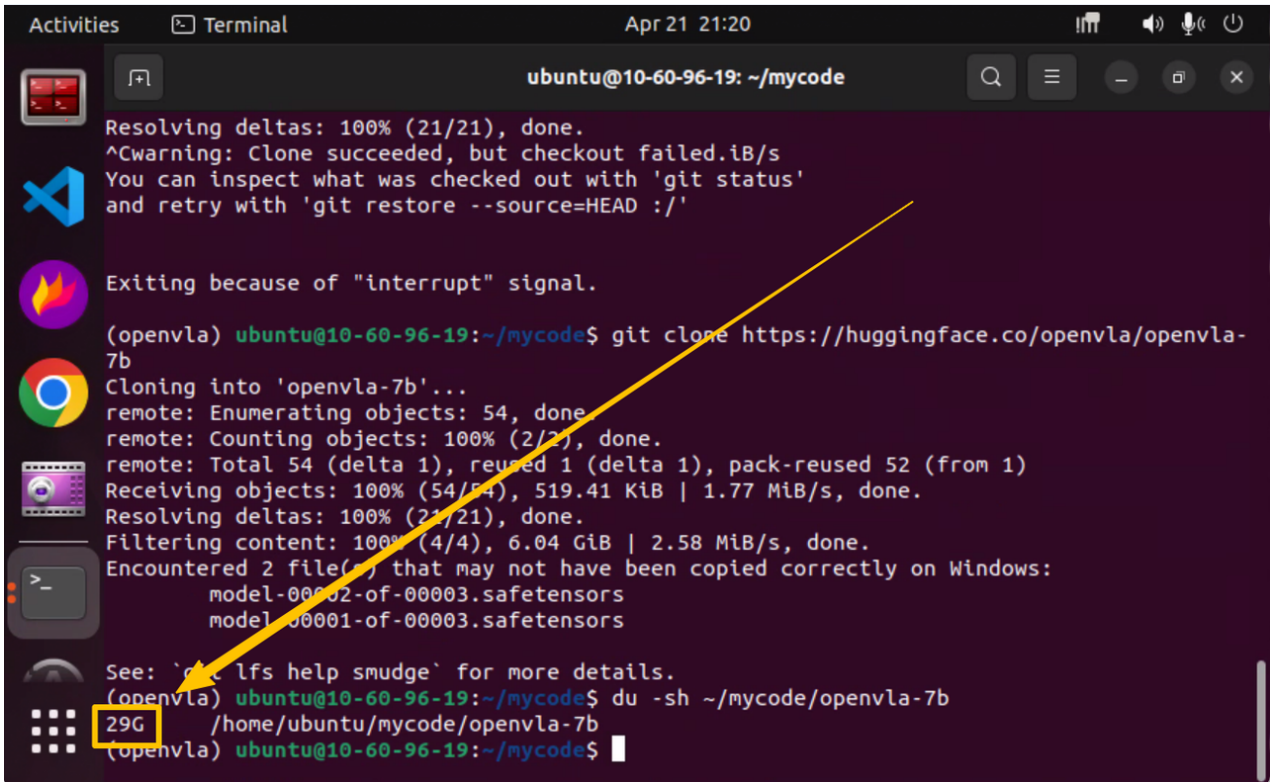
代码块

```
1 apt list --installed | grep nvidia-driver
2 uname -a
3 # Linux 10-60-96-19 5.15.0-153-generic #163-Ubuntu SMP Thu Aug 7 16:37:18 UTC
  2025 x86_64 x86_64 x86_64 GNU/Linux
4
5 nvidia-smi
6
7 cd /home/ubuntu/mycode
```

3.3 克隆仓库（huggingface上的模型）并查看大小（但这个原生模型，要复现libero-spatial中的效果，直接下载fine-tunning后的模型）

原生

```
1 git clone https://huggingface.co/opencvla/opencvla-7b
2 du -sh ~/mycode/opencvla-7b
3 # 29G /home/ubuntu/mycode/opencvla-7b
```



A terminal window titled 'Terminal' with the date 'Apr 21 21:20' and the user 'ubuntu@10-60-96-19: ~/mycode'. The terminal shows the output of a 'git clone' command, which includes a warning about a failed checkout and a message about inspecting the checkout with 'git status'. It then shows the output of a 'du -sh' command, which returns '29G /home/ubuntu/mycode/opencvla-7b'. A yellow arrow points from the '29G' in the 'du' output to the '29G' in the code block above. The terminal also shows the output of a 'git clone' command, which includes a warning about a failed checkout and a message about inspecting the checkout with 'git status'.

```
Resolving deltas: 100% (21/21), done.
^Cwarning: Clone succeeded, but checkout failed.iB/s
You can inspect what was checked out with 'git status'
and retry with 'git restore --source=HEAD :/'

Exiting because of "interrupt" signal.

(openvla) ubuntu@10-60-96-19:~/mycode$ git clone https://huggingface.co/opencvla/opencvla-7b
Cloning into 'opencvla-7b'...
remote: Enumerating objects: 54, done.
remote: Counting objects: 100% (2/2), done.
remote: Total 54 (delta 1), reused 1 (delta 1), pack-reused 52 (from 1)
Receiving objects: 100% (54/54), 519.41 KiB | 1.77 MiB/s, done.
Resolving deltas: 100% (21/21), done.
Filtering content: 100% (4/4), 6.04 GiB | 2.58 MiB/s, done.
Encountered 2 file(s) that may not have been copied correctly on Windows:
    model-00002-of-00003.safetensors
    model-00001-of-00003.safetensors
See: 'git lfs help smudge' for more details.
(openvla) ubuntu@10-60-96-19:~/mycode$ du -sh ~/mycode/opencvla-7b
29G /home/ubuntu/mycode/opencvla-7b
(openvla) ubuntu@10-60-96-19:~/mycode$
```

3.4 配置环境

代码块

```
1 conda create -n openvla python=3.10 -y
2 conda activate openvla
3 conda install pytorch torchvision torchaudio pytorch-cuda=12.1 -c pytorch -c
  nvidia -y
4 conda env list
5 conda list
```

3.5 克隆项目并按readme安装

代码块

```
1 git clone https://github.com/opencvla/opencvla.git
2 cd openvla
3 pip install -e .
4
5 # 安装flash attention
6 pip install packaging ninja
7 ninja --version; echo $? # Verify Ninja --> should return exit code "0"
8 pip install "flash-attn==2.5.5" --no-build-isolation
```

```
9
10 pip list | grep flash-attn
11 # flash-attn                2.5.5
```

3.6 LIBERO环境安装

代码块

```
1 git clone https://github.com/Lifelong-Robot-Learning/LIBERO.git
2 cd LIBERO
3 pip install -e .
4
5 # 下载LIBERO环境所需的依赖
6 cd openvla
7 pip install -r experiments/robot/libero/libero_requirements.txt
8 pip list | grep libero
9 # libero                    0.1.0
```

```
Receiving objects: 100% (489/489), 346.98 KiB | 1.63 MiB/s, done.
Resolving deltas: 100% (194/194), done.
(openvla) ubuntu@10-60-96-19:~/mycode$ cd openvla
(openvla) ubuntu@10-60-96-19:~/mycode/openvla$ pip list | grep flash-attn
Flash-attn                2.5.5
(openvla) ubuntu@10-60-96-19:~/mycode/openvla$ pip list | grep libero
libero                    0.1.0
```

3.7 复现libero-spatial中的效果

代码块

```
1 # 克隆仓库（经原生模型微调后的模型）
2 git clone https://huggingface.co/openvla/openvla-7b-finetuned-libero-spatial
3 du -sh ~/mycode/openvla-7b-finetuned-libero-spatial
```



LIBERO

LIBERO环境



openvla

openvla环境



openvla-7b

原生模型

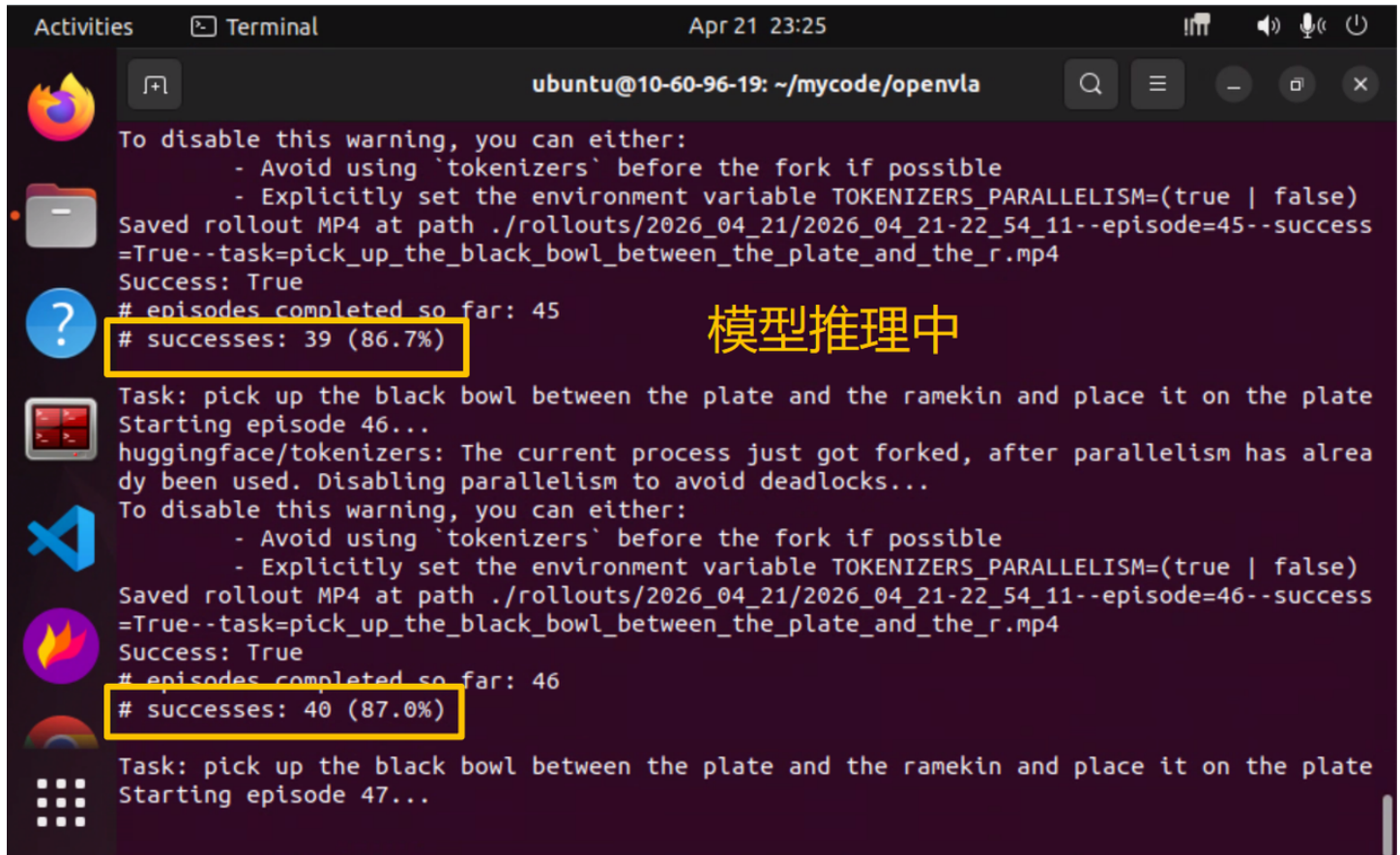


openvla-7b-
finetuned-
libero-sp...

微调模型

```
python experiments/robot/libero/run_libero_eval.py \
2  --model_family openvla \
3  --pretrained_checkpoint /home/ubuntu/mycode/openvla-7b-finetuned-libero-
   spatial/ \
4  --task_suite_name libero_spatial \
5  --center_crop True
```

复现结果：



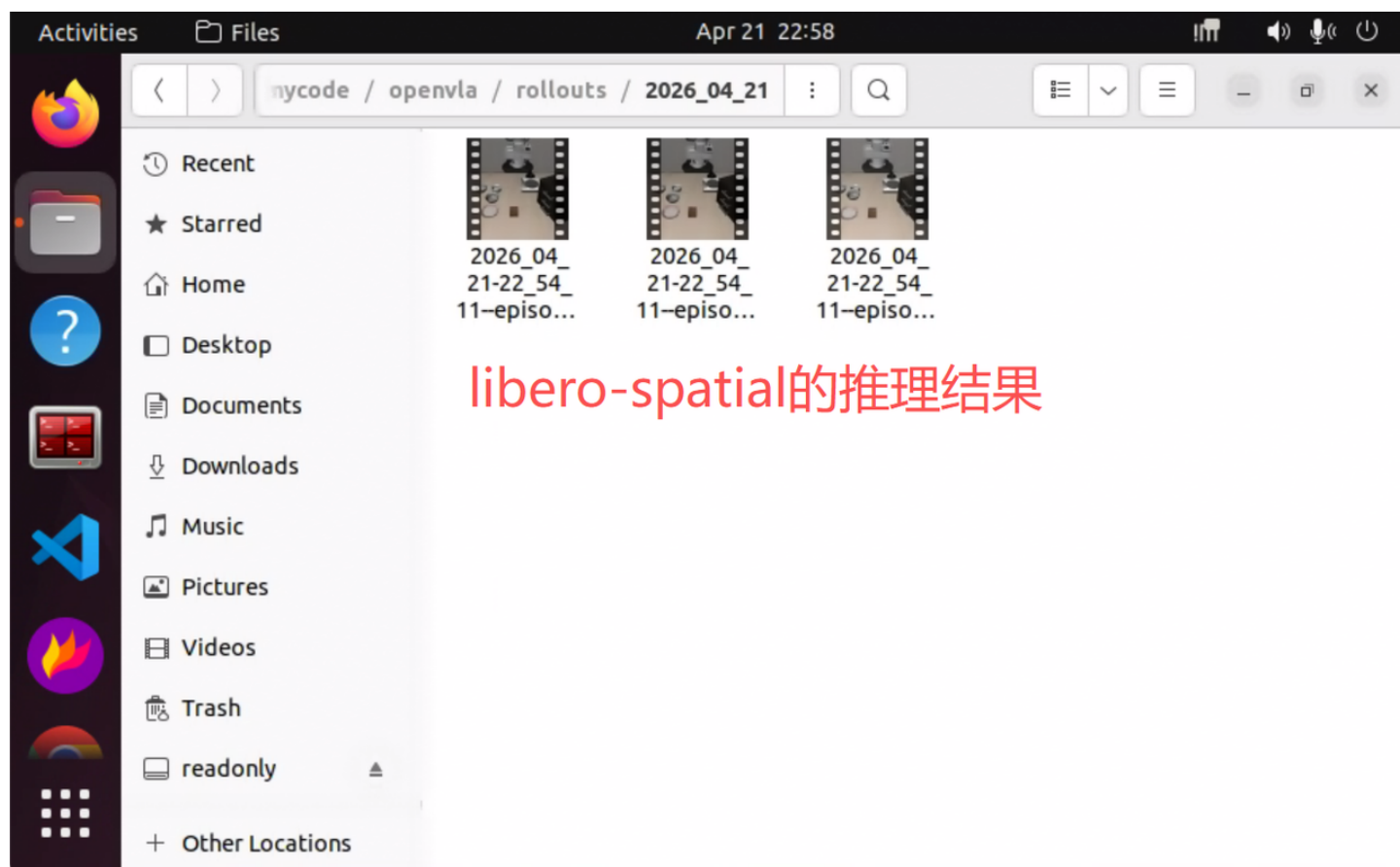
```
Activities Terminal Apr 21 23:25
ubuntu@10-60-96-19: ~/mycode/openvla

To disable this warning, you can either:
  - Avoid using `tokenizers` before the fork if possible
  - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true | false)
Saved rollout MP4 at path ./rollouts/2026_04_21/2026_04_21-22_54_11--episode=45--success
=True--task=pick_up_the_black_bowl_between_the_plate_and_the_r.mp4
Success: True
# episodes completed so far: 45
# successes: 39 (86.7%)

Task: pick up the black bowl between the plate and the ramekin and place it on the plate
Starting episode 46...
huggingface/tokenizers: The current process just got forked, after parallelism has alrea
dy been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
  - Avoid using `tokenizers` before the fork if possible
  - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true | false)
Saved rollout MP4 at path ./rollouts/2026_04_21/2026_04_21-22_54_11--episode=46--success
=True--task=pick_up_the_black_bowl_between_the_plate_and_the_r.mp4
Success: True
# episodes completed so far: 46
# successes: 40 (87.0%)

Task: pick up the black bowl between the plate and the ramekin and place it on the plate
Starting episode 47...
```

模型推理中



- task:pick_up_the_black_bowl_between_the_plate_and_the_r



4. UnifoLM-VLA-0复现

未完待续。。。